

LESSON 5-5 • SESSION 1 • GUIDED NOTES**Volume of Rectangular Prisms**

Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height. · 6.G.A.2

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
volume <i>volumen</i>	The amount of _____ inside a solid.	$V = l \times w \times h$
unit cube <i>cubo unitario</i>	A cube with every edge ____ unit long.	1 cm^3
base area <i>área de la base</i>	The area of the _____ face of a prism.	$B = l \times w$
cubic units <i>unidades cúbicas</i>	Units used to measure _____.	$\text{cm}^3, \text{in}^3, \text{ft}^3$

KEY IDEA*complete each sentence*

- One layer of cubes holds length $\times$ _____ cubes.
- The height tells how many _____ stack up.
- $V = l \times w \times h$, or $V = \underline{\hspace{1cm}} \times h$.

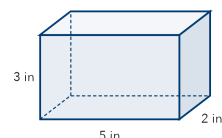
WORKED EXAMPLE*fill in the blanks as we solve*

A box is 6 cm long, 3 cm wide, and 4 cm tall. Find its volume.

Step 1 One layer: $6 \times 3 = \underline{\hspace{1cm}}$ cubes.**Step 2** The height is 4, so there are ____ layers.**Step 3** Multiply: $18 \times 4 = \underline{\hspace{1cm}}$.**Answer** $V = \underline{\hspace{1cm}} \text{ cm}^3$ **YOU TRY***use the example as a model*

1. Find the volume of the prism.

Show your work.

Answer _____ in^3 

2. Find each volume.

a. $10 \text{ m} \times 2 \text{ m} \times 3 \text{ m} \underline{\hspace{1cm}} \text{ m}^3$

b. $B = 25 \text{ ft}^2, h = 4 \text{ ft} \underline{\hspace{1cm}} \text{ ft}^3$

SUM IT UP*in your own words*

Volume uses cubic units because ____.

LESSON 5-5 • SESSION 2 • GUIDED NOTES

Missing Fractional Edges

Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths. • 6.G.A.2

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
edge <i>arista</i>	A _____ where two faces of a solid meet.	length, width, height
base area <i>área de la base</i>	The _____ of the length and width of the base.	$B = l \times w$
inverse operations <i>operaciones inversas</i>	Operations that _____ each other.	$\div$ undoes $\times$

KEY IDEA*complete each sentence*

- $V = l \times w \times h$ works with _____ too.
- To find a missing edge, _____ the volume by the product of the other two edges.
- Check: the three edges should _____ to the volume.

WORKED EXAMPLE*fill in the blanks as we solve*A box holds 72 cm^3 . It is 6 cm long and 4 cm wide. Find its height.**Step 1** Known edges: $6 \times 4 = \underline{\hspace{1cm}} \text{ cm}^2$.**Step 2** Divide: $72 \div 24 = \underline{\hspace{1cm}}$.**Step 3** Check: $6 \times 4 \times 3 = \underline{\hspace{1cm}}$.**Answer** height = cm**YOU TRY***use the example as a model*

1. Find the volume of the prism.

Show your work.

**Answer** _____ in^3

2. Find each value.

a. $V = 60 \text{ m}^3$, base 3 m by 4 m; $h = \underline{\hspace{1cm}} \text{ m}$ **b.** V of $1 \frac{1}{2} \text{ ft} \times 2 \text{ ft} \times 3 \text{ ft}$ _____ ft^3

SUM IT UP*in your own words*

To find a missing edge, I _____ because _____.
